

Bill of Materials (BOM) — Power Tool Battery Cell

Case ID: t3_r1_mimo

Document Number: VBF-T3R1MIMO-BOM-001

Date: 2026-08-30

Cell-Level BOM

Item #	Component	Material	Quantity	Unit Mass (g)	Total Mass (g)
1	Positive electrode (active)	NMC811	1	6.62	6.62
2	Negative electrode (active)	Graphite	1	4.39	4.39
3	Positive current collector	Aluminum (16 μm)	1	4.32	4.32
4	Negative current collector	Copper (12 μm)	1	10.75	10.75
5	Separator	PE (12 μm)	1	0.24	0.24
6	Electrolyte	EC/EMC + LiPF6	~2 mL	~2.4 (est.)	~2.4
7	Cell housing	Al pouch	1	~3.0 (est.)	~3.0
8	Tabs (pos + neg)	Al + Ni	2	~0.5 (est.)	~1.0
	Total cell mass				**~32.7 g**

Notes

- Items 1-5 are from contract-caliber simulation (electrolyte excluded from simulation mass)
- Items 6-8 are domain estimates (not simulated)
- Electrolyte fill volume ≈ pore volume × fill factor: $(35\mu\text{m}\times0.42 + 39\mu\text{m}\times0.32 + 12\mu\text{m}\times0.50) \times 0.1027 \text{ m}^2 \times \sim1.2 \text{ g/cm}^3 \approx$ estimated
- BOM is per single cell; pack-level BOM requires additional BMS, housing, wiring

Material Specifications (Key Items)

Component	Specification	Notes
NMC811 cathode	LiNi $\frac{1}{3}$ Mn $\frac{1}{3}$ Co $\frac{1}{3}$ O $\frac{2}{3}$, D50 ≤ 2 μm	Particle size override from baseline 5.22 μm
Graphite anode	Natural/synthetic graphite, D50 ≤ 2 μm	Particle size override from baseline 5.86 μm
Electrolyte	EC/EMC (3:7) + 1M LiPF6 + additives	Enhanced transport: σ=1.5 S/m, t $_{\text{Li}^+}$ =0.45
Separator	PE, 12 μm, porosity 50%	Shutdown temperature ~130°C
Positive CC	Al, 16 μm	Standard
Negative CC	Cu, 12 μm	Standard